size_balanced

equity strategy | family: equity-baseline | status: validated | derived from `extreme_decile`

What this strategy does

Extreme deciles, size-quintile neutral each quarter

Why it should work. PEAD is stronger in small caps; balancing by size keeps the edge but removes a size bet.

Design notes (from the strategy's source file).

Strategy 3 -- Size-balanced extreme decile.

Strategy 1's trade, but each leg's budget is spread evenly over the size quintiles present that quarter, so micro-caps (where most extreme surprises happen) cannot dominate the book.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
beta_overlay	0.0

Performance

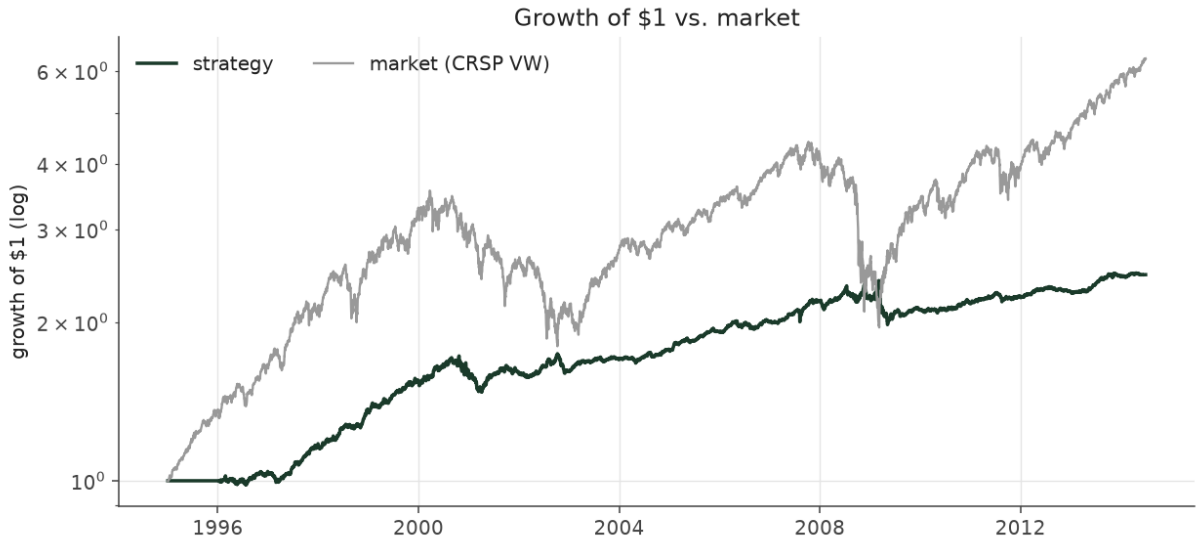
metric	size balanced	market (CRSP VW)
Annualized return	4.74%	9.94%
Annualized volatility	5.26%	19.35%
Sharpe (NAV-pct)	0.91	0.59
Sortino	1.26	
Max drawdown	-17.5%	-55.5%
Calmar	0.27	
Final NAV ($10M start)	$24,622,256	
Worst day	-4.30%	
% positive days	52.2%	
Sharpe 1996-2001	1.32	
Sharpe 2002-2007	1.28	
Sharpe 2008-2013	0.34	

Correlation of daily returns with the market: **0.00**.

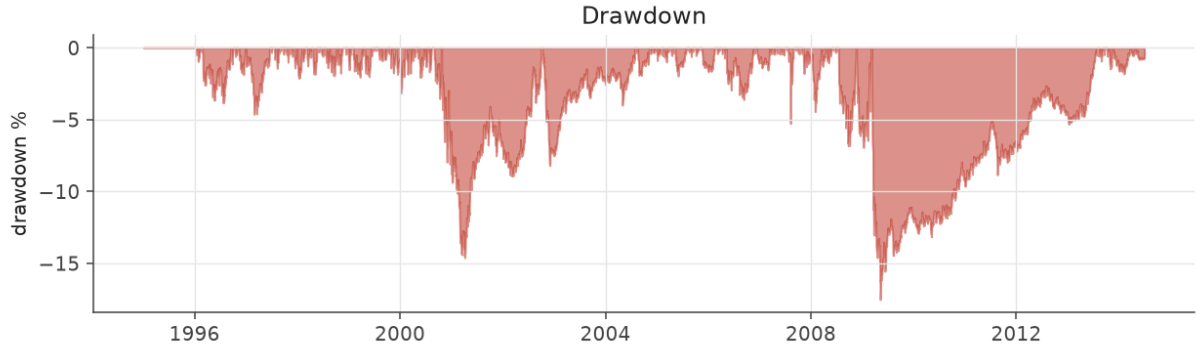
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 0.906

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0474

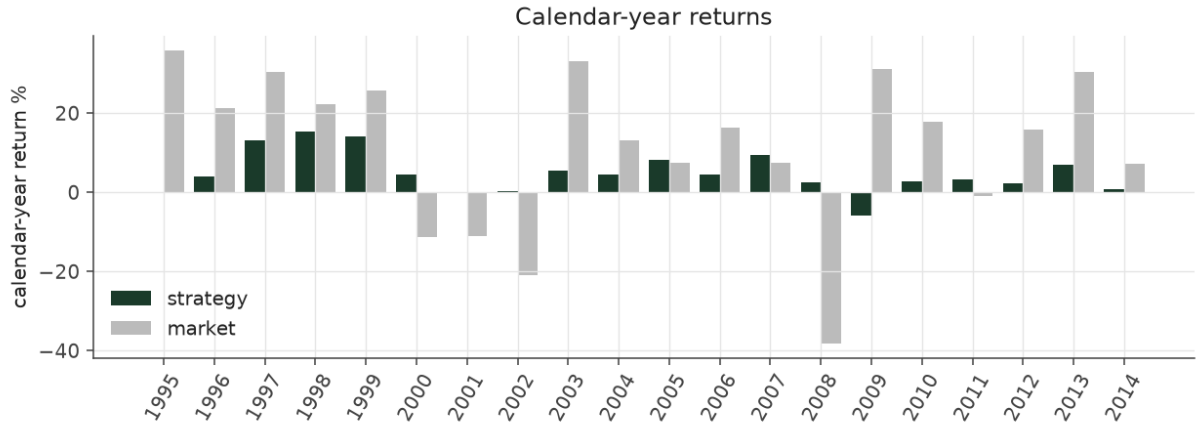
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.27



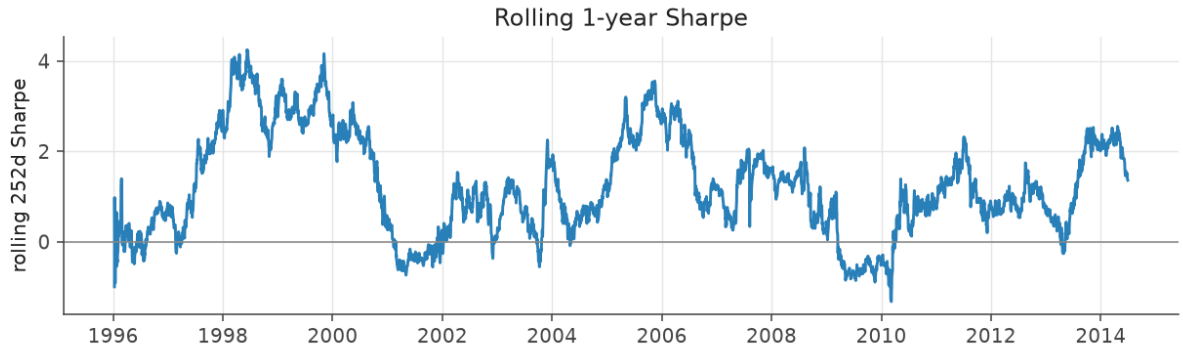
NAV (log scale)



Drawdown from running peak

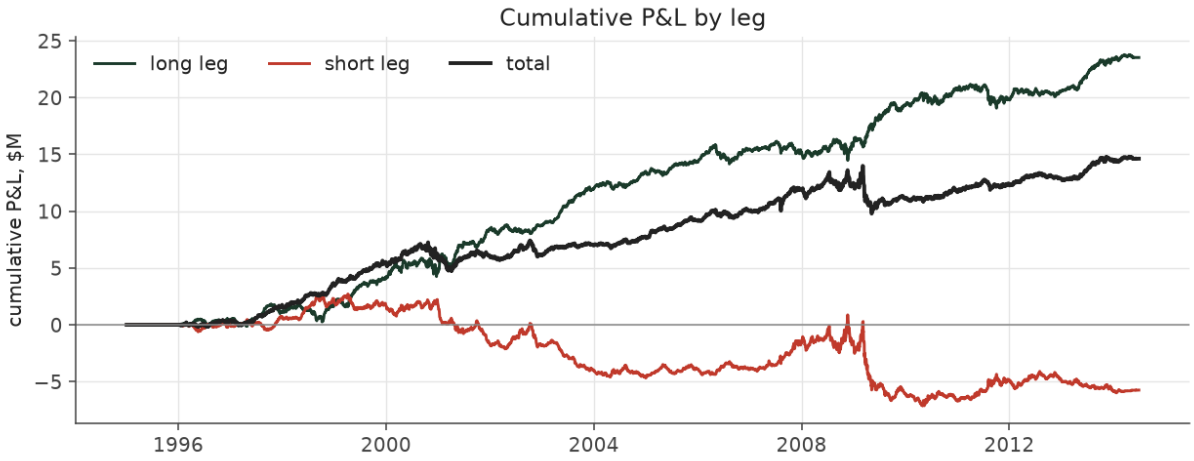


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	23.53	-5.75
P&L after costs ($M)	22.05	-7.43
of which common (average-event) component ($M)	12.07	-12.69
of which excess / PEAD component ($M)	11.46	6.95
excess minus costs ($M)	9.98	5.26
gross return per $ traded (bp)	309.93	-70.95
positions in the money (hit rate)	0.57	0.49
stand-alone Sharpe (leg only)	1.16	-0.23

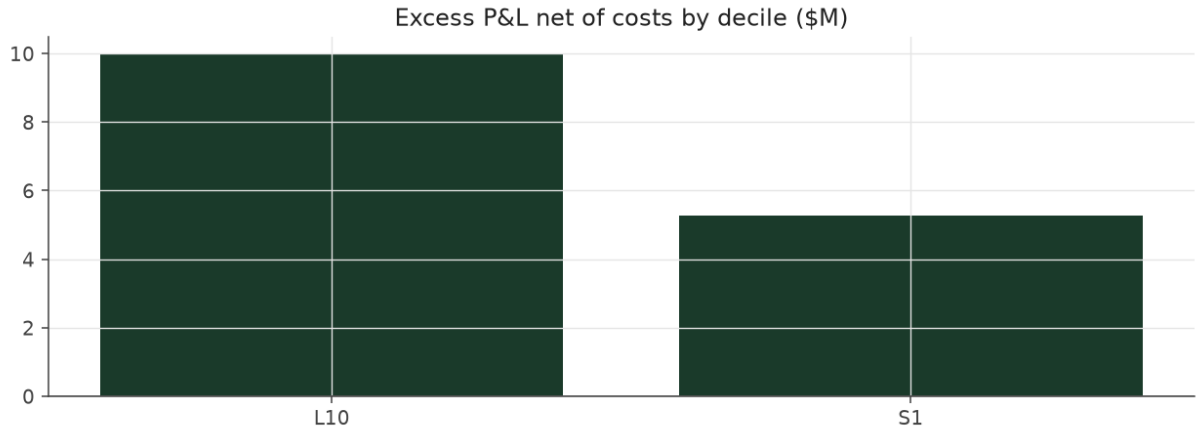
Daily P&L correlation between the two legs: **-0.64**.

• **The short leg loses money in absolute terms** (-5.7M before costs) while the long leg earns +23.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 12.1M / 12.7M. Stripping that common component, the surprise signal itself earned +11.5M on the long side and +6.9M on the short side.

• The two legs' daily P&L are -0.64 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.16 long / -0.23 short vs 0.91 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	17,174	759.10	22.05	11.46	1.48	9.98	65.5%	131.48
short	1	17,169	809.89	-7.43	6.95	1.68	5.26	34.5%	64.99



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,832	149.12	2.16	3.31	0.54	2.77	18.2%	185.60
2	7,159	327.44	4.51	5.49	0.90	4.59	30.1%	140.12
3	7,735	358.82	5.24	5.93	0.74	5.19	34.0%	144.64
4	7,731	365.93	1.70	2.26	0.57	1.69	11.1%	46.23
5	6,886	367.69	1.01	1.42	0.41	1.01	6.6%	27.39

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	4,357	93.37	2.65	3.33	0.47	2.86	18.8%	306.58
2	7,223	337.61	3.47	5.06	1.01	4.05	26.6%	119.93
3	7,834	386.64	5.77	6.76	0.77	5.99	39.3%	154.91
4	7,850	385.18	1.56	2.11	0.54	1.57	10.3%	40.68
5	7,077	366.08	1.20	1.17	0.37	0.80	5.2%	21.83

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	6,338	335.10	-0.26	3.39	0.51	2.88	18.9%	85.99
41-60d	11,202	486.79	1.49	2.35	0.99	1.36	8.9%	27.93

61-90d	12,665	583.62	5.83	5.09	1.17	3.92	25.7%	67.14
>90d	4,138	163.48	7.56	7.57	0.48	7.08	46.5%	433.38

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	9,920	371.88	5.68	6.62	0.71	5.92	38.8%	159.05
2002-07	11,506	527.09	5.64	6.77	1.04	5.73	37.6%	108.64
2008-13	12,917	670.02	3.30	5.01	1.41	3.60	23.6%	53.78

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	6,718	298.31	2.57	1.93	0.59	1.33	8.7%	44.67
Chems	687	31.76	-0.19	-0.14	0.06	-0.20	-1.3%	-61.42
Durbl	869	38.28	0.26	0.51	0.08	0.43	2.8%	111.43
Enrgy	1,375	66.03	1.65	1.81	0.13	1.68	11.0%	254.65
Hlth	3,037	133.90	1.35	1.18	0.29	0.89	5.8%	66.46
Manuf	3,627	163.12	0.76	1.50	0.32	1.18	7.8%	72.50
Money	5,366	255.16	1.65	3.64	0.50	3.14	20.6%	123.09
NoDur	1,540	71.33	0.94	1.05	0.14	0.91	6.0%	127.65
Other	5,074	228.66	2.19	2.65	0.51	2.14	14.1%	93.69
Shops	3,547	160.84	2.53	2.91	0.32	2.59	17.0%	160.85
Telcm	1,164	55.57	0.74	1.00	0.11	0.89	5.8%	160.29
Utils	1,339	66.02	0.17	0.36	0.11	0.26	1.7%	38.57

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	11,517	531.47	2.60	4.04	0.96	3.08	20.2%	57.99
low vol	11,554	516.11	6.39	7.84	1.15	6.70	44.0%	129.83
mid vol	10,730	492.43	5.30	6.13	0.99	5.14	33.7%	104.34

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	4.74%	0.00%
0.5%	0.9	4.53%	0.20%
1.0%	1.8	4.32%	0.41%
2.0%	3.6	3.88%	0.85%
5.0%	9.1	2.29%	2.44%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.